REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

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Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 12 Advanced | Module 22 Lesson 2 | Matter Waves

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: $\lambda = h/p = h/(mv)$ for nonrelativistic particles; $\Delta x \Delta p \geq h/(4\pi)$. $h = 6.626 \times 10^{-34}$ J s; $m_e = 9.11 \times 10^{-31}$ kg.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 12 book, printed pages 59–61.

P01. The de Broglie wavelength of a particle is:

A. hp B. p/h
C. h/p D. mc^2

Answer

C. h/p.

P02. If particle momentum doubles, its de Broglie wavelength:

A. Doubles B. Quadruples
C. Stays unchanged D. Halves

Answer

D. Halves.

P03. Electron diffraction demonstrates electrons' :

A. Wave behaviour B. Zero momentum
C. Lack of mass D. Classical paths only

Answer

A. Wave behaviour.

P04. The uncertainty principle concerns a fundamental limit on simultaneous knowledge of:

A. Mass and charge only B. Position and momentum
C. Wavelength and colour only D. Energy and mass in every measurement

Answer

B. Position and momentum.

P05. Macroscopic objects normally show negligible diffraction because their de Broglie wavelengths are:

A. Larger than the room B. Always exactly zero
C. Extremely small compared with relevant openings D. Independent of momentum

Answer

C. Extremely small compared with relevant openings.

P06. A particle has momentum 2×10^{-24} kg m/s. Find its de Broglie wavelength. Use $h = 6.626 \times 10^{-34}$ J s.

A. 3.31×10^{-10} m B. 6.63×10^{-10} m
C. 1.66×10^{-10} m D. -3.31×10^{-10} m

Answer

A. 3.31×10^{-10} m. $\lambda = h/p = (6.626 \times 10^{-34})/(2 \times 10^{-24}) = 3.31 \times 10^{-10}$ m.

P07. An electron moves at 2×10^{6} m/s. Find its nonrelativistic de Broglie wavelength. Use $m_e = 9.11 \times 10^{-31}$ kg and $h = 6.626 \times 10^{-34}$ J s.

A. 7.27×10^{-10} m B. 3.64×10^{-10} m
C. 1.82×10^{-10} m D. -3.64×10^{-10} m

Answer

B. 3.64×10^{-10} m. $\lambda = h/(mv) = (6.626 \times 10^{-34})/[(9.11 \times 10^{-31})(2 \times 10^{6})] = 3.64 \times 10^{-10}$ m.

P08. A particle has de Broglie wavelength 2×10^{-10} m. Find its momentum. Use $h = 6.626 \times 10^{-34}$ J s.

A. 6.63×10^{-24} kg m/s B. 1.66×10^{-24} kg m/s
C. 3.31×10^{-24} kg m/s D. -3.31×10^{-24} kg m/s

Answer

C. 3.31×10^{-24} kg m/s. $p = h/\lambda = (6.626 \times 10^{-34})/(2 \times 10^{-10}) = 3.31 \times 10^{-24}$ kg m/s.

P09. The momentum of a particle is multiplied by 3. Calculate $\lambda_{new}/\lambda_{old}$ for its de Broglie wavelength.

A. 0.6667 B. 0.1667
C. -0.3333 D. 0.3333

Answer

D. 0.3333. $\lambda = h/p$, so $\lambda_{new}/\lambda_{old} = 1/3 = 0.3333$.

P10. Use $\Delta x \Delta p \geq h/(4\pi)$ to find the minimum momentum uncertainty for $\Delta x = 2 \times 10^{-10}$ m. Use $h = 6.626 \times 10^{-34}$ J s.

A. 2.64×10^{-25} kg m/s B. 5.27×10^{-25} kg m/s
C. 1.32×10^{-25} kg m/s D. -2.64×10^{-25} kg m/s

Answer

A. 2.64×10^{-25} kg m/s. $\Delta p_{min} = h/(4\pi\Delta x) = (6.626 \times 10^{-34})/(4\pi \times 2 \times 10^{-10}) = 2.64 \times 10^{-25}$ kg m/s.

P11. State de Broglie's hypothesis.

Answer

Particles of matter can exhibit wave properties with wavelength $\lambda = h/p$.

P12. Why is electron diffraction evidence for matter waves?

Answer

An interference/diffraction pattern is a characteristic wave phenomenon, showing that electron behaviour cannot be described solely by classical particles.

P13. State Heisenberg's position–momentum uncertainty principle.

Answer

The uncertainties satisfy $\Delta x \Delta p \geq h/(4\pi)$, so position and momentum cannot both be specified arbitrarily precisely.

P14. How does increasing speed affect the de Broglie wavelength of the same nonrelativistic particle?

Answer

Momentum mv increases, so wavelength h/(mv) decreases.

P15. Does the uncertainty principle mean only that instruments are poorly made?

Answer

No. It reflects a fundamental property of quantum states, not merely avoidable experimental defects.

P16. A particle has momentum 3×10^{-24} kg m/s. Find its de Broglie wavelength. Use $h = 6.626 \times 10^{-34}$ J s.

Answer

$\lambda = h/p = (6.626 \times 10^{-34})/(3 \times 10^{-24}) = 2.21 \times 10^{-10}$ m.

P17. A particle has momentum 4×10^{-24} kg m/s. Find its de Broglie wavelength. Use $h = 6.626 \times 10^{-34}$ J s.

Answer

$\lambda = h/p = (6.626 \times 10^{-34})/(4 \times 10^{-24}) = 1.66 \times 10^{-10}$ m.

P18. An electron moves at 3×10^{6} m/s. Find its nonrelativistic de Broglie wavelength. Use $m_e = 9.11 \times 10^{-31}$ kg and $h = 6.626 \times 10^{-34}$ J s.

Answer

$\lambda = h/(mv) = (6.626 \times 10^{-34})/[(9.11 \times 10^{-31})(3 \times 10^{6})] = 2.42 \times 10^{-10}$ m.

P19. An electron moves at 4×10^{6} m/s. Find its nonrelativistic de Broglie wavelength. Use $m_e = 9.11 \times 10^{-31}$ kg and $h = 6.626 \times 10^{-34}$ J s.

Answer

$\lambda = h/(mv) = (6.626 \times 10^{-34})/[(9.11 \times 10^{-31})(4 \times 10^{6})] = 1.82 \times 10^{-10}$ m.

P20. A particle has de Broglie wavelength 3×10^{-10} m. Find its momentum. Use $h = 6.626 \times 10^{-34}$ J s.

Answer

$p = h/\lambda = (6.626 \times 10^{-34})/(3 \times 10^{-10}) = 2.21 \times 10^{-24}$ kg m/s.

P21. A particle has de Broglie wavelength 4×10^{-10} m. Find its momentum. Use $h = 6.626 \times 10^{-34}$ J s.

Answer

$p = h/\lambda = (6.626 \times 10^{-34})/(4 \times 10^{-10}) = 1.66 \times 10^{-24}$ kg m/s.

P22. The momentum of a particle is multiplied by 4. Calculate $\lambda_{new}/\lambda_{old}$ for its de Broglie wavelength.

Answer

$\lambda = h/p$, so $\lambda_{new}/\lambda_{old} = 1/4 = 0.25$.

P23. The momentum of a particle is multiplied by 5. Calculate $\lambda_{new}/\lambda_{old}$ for its de Broglie wavelength.

Answer

$\lambda = h/p$, so $\lambda_{new}/\lambda_{old} = 1/5 = 0.2$.

P24. Use $\Delta x \Delta p \geq h/(4\pi)$ to find the minimum momentum uncertainty for $\Delta x = 3 \times 10^{-10}$ m. Use $h = 6.626 \times 10^{-34}$ J s.

Answer

$\Delta p_{min} = h/(4\pi\Delta x) = (6.626 \times 10^{-34})/(4\pi \times 3 \times 10^{-10}) = 1.76 \times 10^{-25}$ kg m/s.

P25. Use $\Delta x \Delta p \geq h/(4\pi)$ to find the minimum momentum uncertainty for $\Delta x = 4 \times 10^{-10}$ m. Use $h = 6.626 \times 10^{-34}$ J s.

Answer

$\Delta p_{min} = h/(4\pi\Delta x) = (6.626 \times 10^{-34})/(4\pi \times 4 \times 10^{-10}) = 1.32 \times 10^{-25}$ kg m/s.

P26. An electron and a proton have equal momentum. Compare their de Broglie wavelengths.

Answer

They are equal because $\lambda = h/p$ depends on momentum, not on mass separately.

P27. An electron and a proton have equal nonrelativistic speeds. Which has the longer wavelength?

Answer

The electron has much smaller mass and momentum at the same speed, so it has the longer de Broglie wavelength.

P28. If position uncertainty is reduced to one quarter, how does the minimum momentum uncertainty change?

Answer

It increases by a factor of four under the uncertainty-product limit, assuming comparison at that bound.

P29. Why can a crystal diffract electrons?

Answer

Its regular atomic spacing provides a periodic structure on a scale comparable with suitable electron de Broglie wavelengths, producing interference and diffraction.

P30. Explain why treating an electron as a tiny ball on a precisely known trajectory is limited.

Answer

Quantum states have wave properties and a fundamental position–momentum uncertainty. A classical exact trajectory cannot fully represent their behaviour.